Ali Tosyali, PhD

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EMPLOYMENT

Assistant Professor of Management Information Systems *Aug 2019 - Present*
University of Delaware – Accounting and MIS

Lecturer *Jan 2017 – Jun 2019*
Rutgers Business School – Management Science and Information Systems

EDUCATION

Rutgers, The State University of New Jersey *Piscataway, NJ*
Ph.D. in Industrial & Systems Engineering *2019*

- Title: Development of advanced data mining algorithms for the analysis of directed networks
- Committee: Dr. Myong K. Jeong (chair), Dr. Susan Albin, Dr. Hoang Pham, Dr. Jie Gong, and Dr. Weihong Guo

M.S. in Industrial & Systems Engineering *2016*

Turkish Military Academy *Ankara, Turkey*
B.S. in Business Administration and Management *2013*

RESEARCH INTERESTS

Method Area: Network Science, Machine Learning, Complex Systems
Application Area: Business Analytics, Financial Services Analytics, Social Network Analysis, Opinion Mining and Analysis

PUBLICATIONS

Journals (published/accepted)

1. A novel method for identifying competitors using a financial transaction network.
   - J. Choi, A. Tosyali, B. Kim, H. Lee, and MK. Jeong.
   - ***IEEE Transactions on Engineering Management*** (forthcoming)
2. Data-driven gantry health monitoring and process status identification by utilizing texture extraction
   - A. Tosyali, R. Song, W. Guo, A. Abolhassani, and R. Kalamdani.
   - ***Journal of Computing and Information Science in Engineering*** 21.1 (2021)
3. New node anomaly detection algorithm based on nonnegative matrix factorization for directed citation networks.
   - A. Tosyali, J. Kim, J. Choi, Y. Kang, and MK. Jeong.
   - ***Annals of Operations Research*** 288 (2020): 450-474.
4. Regularized asymmetric nonnegative matrix factorization for clustering in directed networks.
   - A. Tosyali, J. Kim, J. Choi, and MK. Jeong.
   - ***Pattern Recognition Letters*** 125 (2019): 750-757.
5. Patent clustering and outlier ranking methodologies for attributed patent citation networks for technology opportunity discovery.
   - A. Rodriquez, A. Tosyali, B. Kim, J. Choi, J. Lee, B. Coh, and MK. Jeong.
   - ***IEEE Transactions on Engineering Management*** 63.4 (2016): 426-437.

6. As a supply chain financing source, trade credit and bank credit relationship during financial crises from clustering point of view.
   - C. Sevim, A. Ekiyor, and A. Tosyali.
   - ***International Business Research*** 9.4 (2016): 45-57.

Journals (submitted/under review)

7. A novel similarity-based link prediction approach for transaction networks
8. A dynamic graph-based approach to ranking firms for identifying key players using inter-firm transactions
9. Intelligent energy framework for forecasting volatile electricity consumption behavior: The case of Turkey
10. A node-based index for clustering validation of graph data

Working Papers

11. Influential patent identification in patent citation networks
12. Community effect on persuader identification in digital social networks
13. Online fake review detection using network structure and semantics
14. Bayesian asymmetric nonnegative matrix factorization

Refereed Conference Proceedings

15. B. Tavakkol and A. Tosyali. Density-based clustering validation of uncertain data objects. In *2020 Northeast Decision Sciences Institute Annual Meeting*.
16. M. Usta and A. Tosyali. Characterization of model-based uncertainties in incompressible turbulent flows by machine learning. In *ASME 2018 International Mechanical Engineering Congress and Exposition*.

Conference Presentations

17. New Data Mining Models to Analyze Big Value Chain Networks Data. *UKC*, 2017
18. Ranking Outliers in Patent Citation Network Using Attribute and Graph Structure. *INFORMS Annual Meeting*, 2015.

TEACHING

Accounting and Management Information Systems, University of Delaware

- Introduction to Programming Business Applications
- Fundamentals of Business Analytics (*Course coordinator*)
- Social Network Analysis (*Designed the course*)

Management Science and Information Systems, Rutgers Business School

Undergraduate Level Courses

- Business Decision Analytics under Uncertainty
- Operations Management
- Statistical Methods in Business
- Management Information Systems

Graduate Level Courses

- Data Analysis and Decision Making
- Analytics for Business Intelligence

SERVICES

University of Delaware

- Faculty Search Committee Member for Fall 2020 TT Assistant Professor of MIS
- Course coordinator for MISY262: Fundamentals of Business Analytics

Reviewer

Journals

- IEEE Transactions on Knowledge and Data Engineering (TKDE)
- IEEE Transactions on Cybernetics (CYB)
- IEEE Transactions on Engineering Management (TEM)
- Annals of Operations Research (ANOR)
- Journal of Manufacturing Systems (SMEJMS)
- Quality and Reliability Engineering International (QRE)

Conferences

- International Conference on Information Systems (ICIS)

HONORS & AWARDS

2019 – **Excellence in Teaching and Leadership** (nominated), Rutgers University, NJ
2019 – **Outstanding Doctoral Student Award** (nominated), Rutgers University, NJ
2013 – **Science Award**, Turkish Armed Forces, Turkey
2013 – **Full Scholarship**, including tuition and monthly stipend, Ministry of National Defense, Turkey
2012 – **Academic Excellency**, 11th Hwarang Collegiate Forum, South Korea
2012 – **Creativity Award**, 11th Hwarang Collegiate Forum, South Korea

SKILLS

Programming Languages and Tools: Python, R, MATLAB, SQL, LaTeX, Microsoft Excel, yEd, Gephi
Spoken Languages: Turkish (native), English (fluent)
Extracurricular Activities: Racquetball, mountain biking, baglama (a folkloric instrument), electric guitar, snowboarding, soccer